

VENEEL KUMAR A

+91-9606288164 | veneeldas77@gmail.com | linkedin.com/in/veneelkumara77 | github.com/Veneel77 | Bangalore, India

SUMMARY

Python-focused Software Engineer with hands-on experience developing, testing, and maintaining AI-powered applications, REST APIs, backend services, and data processing pipelines. Proficient in Object-Oriented Programming, Data Structures & Algorithms, SQL, Git, and SDLC. Experienced with automation scripting, cloud platforms (AWS, Azure, GCP), FastAPI, and AI/ML application development including chatbot-style RAG systems and agentic AI pipelines. Strong analytical and problem-solving skills with a published research background in deep learning.

SKILLS

Programming Languages: Python, Java, SQL

Software Engineering:

Object-Oriented Programming (OOP), Data Structures & Algorithms, SDLC, Code Review, Testing & Debugging, Version Control (Git)

Backend & APIs:

FastAPI, REST APIs, Microservices Architecture, API Design, CI/CD, Docker, Docker Compose, Automation Scripting

Data & Analytics:

Pandas, NumPy, Apache Spark, PySpark, Airflow, ETL/ELT Pipelines, Data Processing, Reporting (Power BI)

Cloud Platforms:

AWS (SageMaker, EC2, S3, IAM), Azure (AI Services, Functions, App Services), GCP

AI / ML:

Python AI/ML Applications, LLMs (GPT, LLaMA via HuggingFace), LangChain, CrewAI, RAG Architecture, TensorFlow, Keras, Scikit-Learn, Chatbot & Agentic AI Systems

Databases:

MySQL, MongoDB, PostgreSQL, ChromaDB (Vector DB), SQL & NoSQL

Developer Tools:

GitHub, Jupyter Notebook, Streamlit, Power BI, HTML/CSS (familiar)

EXPERIENCE

AI/ML Engineer Trainee · Rooman Technologies · Remote

Sep 2024 – Feb 2025

- Developed, tested, and maintained Python-based ML applications and scripts (classification, regression, ensemble methods) using Scikit-Learn; applied structured debugging, performance tuning, and code review practices.
- Built a Heart Disease Prediction application using Python and ensemble methods (Random Forest, Gradient Boosting) — implemented clean OOP code with modular data preprocessing, feature engineering, and evaluation pipelines.
- Prepared technical documentation and project updates; collaborated with team to adopt SDLC best practices throughout the development lifecycle.

PROJECTS

Agentic Data Pipeline Reliability Platform

Jun 2026 – Present

Python · CrewAI · Google Gemini 2.5 Flash · ChromaDB · FastAPI · Docker | github.com/Veneel77

- Developed and maintained a Python-based AI application automating ETL incident lifecycle management — failure classification, root cause analysis, fix recommendation, and report generation using a 5-agent orchestration system.
- Engineered a production-ready FastAPI REST backend exposing the pipeline as a scalable microservice; containerised with Docker Compose for reproducible deployment; implemented structured logging and observability hooks.
- Built a RAG-powered chatbot-style documentation retrieval module integrating ChromaDB vector store for semantic search over Spark/Airflow knowledge bases — an AI-based automation tool reducing manual investigation time.

RAG-Based Knowledge Assistant

Nov – Dec 2025

Python · LangChain · ChromaDB · HuggingFace · FastAPI · Streamlit | github.com/Veneel77/RAG-based-Knowledge-assistant

- Designed and built a Python AI chatbot application for semantic, document-aware question-answering; developed modular REST API backend (FastAPI) and data ingestion pipeline with clean separation of concerns.
- Integrated HuggingFace sentence-transformer embeddings with ChromaDB for accurate retrieval; implemented unit tests for ingestion and retrieval pipelines; open-sourced under MIT licence with full technical documentation.

Customer Churn Prediction System

Sep 2025

Python · XGBoost · Scikit-Learn · Pandas · NumPy · Power BI | github.com/Veneel77/Customer-Churn-Prediction-System

- Developed a Python data processing and analytics application on IBM Telco dataset (7,000+ records, 21 features) — data cleaning, feature engineering with Pandas/NumPy, ensemble model training (XGBoost, ~85% accuracy).
- Built a Power BI reporting and data visualisation layer for business stakeholders; documented the full pipeline and analysis methodology as a technical report.

RESEARCH & PUBLICATIONS

Alzheimer's Diagnosis via Deep Learning

JETIR, 2025

- Published peer-reviewed research applying Python-based CNN architectures and transfer learning (VGG16, ResNet50) on MRI datasets for multi-class Alzheimer's stage classification; achieved >90% accuracy through hyperparameter tuning and data augmentation. Journal of Emerging Technologies and Innovative Research (JETIR). 2025.